INCH-POUND
MIL-DTL-20625C
23 April 2019
SUPERSEDING
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27 August 2013

DETAIL SPECIFICATION

UNIVERSAL JOINTS, BLOCK TYPE, MECHANICAL REMOTE CONTROL SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers universal joints of the block type for use in rotary type hand-operated or intermittent power-operated valve remote control systems of the mechanical type.

1.2 Classification. Universal joints are of the following classes, as specified (see 6.1):

a. Class A – Steel

b. Class B – Bronze

c. Class C – Steel, Corrosion-Resistant

1.3 Part or identifying number (PIN). PINs to be used to universal joints acquired to this specification (see 6.2) are created as follows:

M	**20625**	-	**X**	**Y**
Prefix for Military Specification	Specification Number		Class (see code below)	Joint Size (see code below)

Class Code	
Class	**Code**
A – Steel	A
B – Bronze	B
C – Steel, Corrosion-Resistant	C

Joint Size Code											
Joint Size	**Code**	**Joint Size**	**Code**	**Joint Size**	**Code**	**Joint Size**	**Code**	**Joint Size**	**Code**	**Joint Size**	**Code**
⅜	1	¾	4	1⅛	7	1¾	10	2½	13	4	16
½	2	⅞	5	1¼	8	2	11	2¾	14		
⅝	3	1	6	1½	9	2¼	12	3	15		

Example: A 2-inch steel universal joint of the block type would be PIN M20625-A11.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3010

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A304	-	Standard Specification for Carbon and Alloy Steel Bars Subject to End-Quench Hardenability Requirements
ASTM A582/A582M	-	Standard Specification for Free-Machining Stainless Steel Bars
ASTM B21/B21M	-	Standard Specification for Naval Brass Rod, Bar, and Shapes
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire

(Copies of these documents are available online at www.astm.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The materials for parts shall be in accordance with table I as applicable. Materials for parts not covered by table I shall be of a high grade commercial standard material that, when installed in a mechanical valve remote control system and operated intermittently, will transmit the required torques in accordance with table II (see 4.1.2).

TABLE I. Materials.

	Hubs	Block	Pins
Class A	ASTM A304	ASTM A304	ASTM A304
Class B	ASTM B21/B21M	ASTM B21/B21M	ASTM B164
Class C	ASTM A582/A582M	ASTM A582/A582M	ASTM A582/A582M

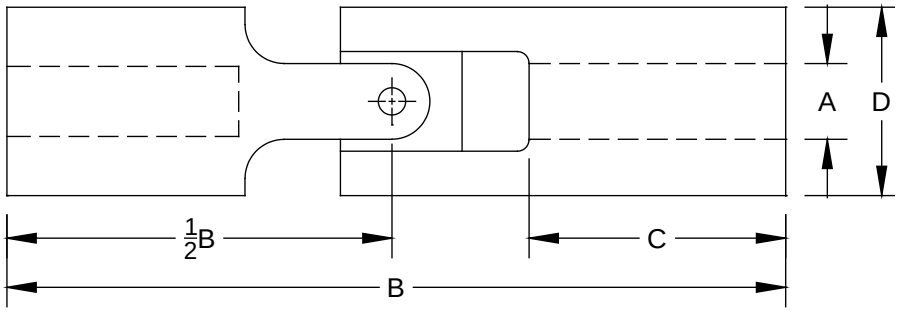
TABLE II. Transmitted torque (foot-pounds).

Joint Size	150 Degrees	180 Degrees
⅜	2	2.5
½	5.6	7
⅝	12	15
¾	14	17.5
⅞	16	20
1	32	40
1⅛	41.6	52
1¼	56	70
1½	68	105
1¾	80	125
2	130	180
2¼	170	215
2½	200	250
2¾	250	310
3	300	375
4	400	500

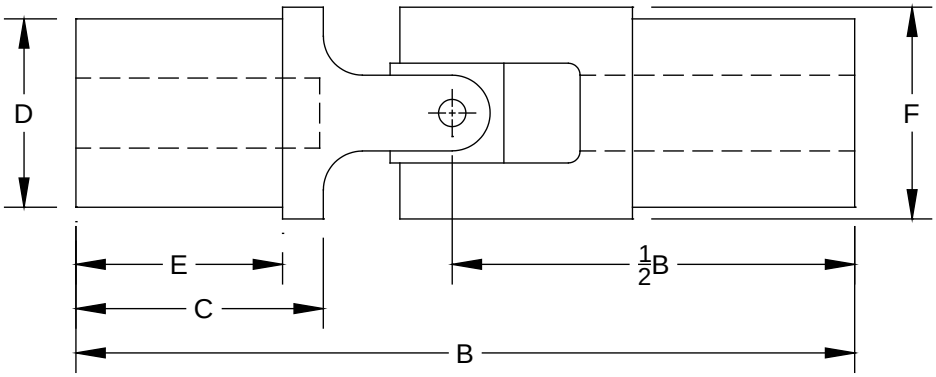
3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 General design. Universal joints shall consist of two hubs with extended jaws, herein called the “yoke,” a block, and bearing pins assembled so that a torque applied to one hub will be transmitted to the other. The dimensions shall be in accordance with table III, and joints shall be assembled in accordance with figure 1 (see 4.2.1).

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CLASSES A AND C



CLASS B

FIGURE 1. Universal joints.

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TABLE III. General dimensions (inches).

A, Shaft Size (max.)	B, Overall Length		C, Hole Length		D, Hub Diameter		E, Hub Length Class B	F, Offset Diameter Class B
	Classes A and C	Class B	Classes A and C	Class B	Classes A and C	Class B		
3/16	1 3/4	4 3/4	11/16	1 1/2	3/8	1 1/4	1 1/8	1 3/8
1/4	2	4 3/4	3/4	1 1/2	1/2	1 1/4	1 1/8	1 3/8
5/16	2 1/4	4 3/4	13/16	1 1/2	5/8	1 1/4	1 1/8	1 3/8
3/8	2 11/16	4 3/4	31/32	1 1/2	3/4	1 1/4	1 1/8	1 3/8
1/2	3	4 3/4	1 1/32	1 1/2	7/8	1 1/4	1 1/8	1 3/8
5/8	3 3/8	4 3/4	1 1/8	1 1/2	1	1 1/4	1 1/8	1 3/8
9/16	3 1/2	4 3/4	1 7/32	1 1/2	1 1/8	1 1/4	1 1/8	1 3/8
3/4	3 3/4	4 3/4	1 3/16	1 1/2	1 1/4	1 1/4	1 1/8	1 3/8
7/8	4 1/4	4 3/4	1 11/32	1 1/2	1 1/2	1 1/4	1 1/8	1 3/8
1	5	6 3/4	1 9/16	2 1/4	1 3/4	1 3/4	1 3/4	2
1 1/4	5 7/16	8	1 5/8	2 1/2	2	2 1/4	1 7/8	2 3/4
1 1/2	7	8	2 5/32	2 1/2	2 1/2	2 1/4	1 7/8	2 3/4
1 3/4	9	9	2 11/16	2 3/4	3	2 3/4	2 1/8	3 1/8
2	10 5/8	----	3 1/8	----	4	----	----	----

3.3.1 Sizes. Universal joints shall be designated by the outside diameter of the hub (see D of table III). For a given shaft size, the joint shall be equal to or larger than that in table III as specified (see 4.2.1).

3.3.2 Hubs. Unless otherwise specified (see 6.2), the joints shall be furnished with the hubs machined with centers (drilled), and the hub dimensions (diameter and length) shall be as specified in table III.

3.3.3 Bearing pins. The bearing pins, when assembled with the hubs and block, shall not project outward beyond the yoke and shall be positively secured in the operating position so that operation will not cause the pins to become loose or drop out (see 4.2.1).

3.3.4 Dimensional tolerances. Clearance permitted between pins and holes shall be a maximum of 0.001 inch. Clearance between the block and yoke shall be a maximum of 0.004 inch. Pins may be off-center a maximum of 0.005 inch. All other dimensions shall be ±0.020 inch (see 4.2.1).

3.3.5 Lubrication. An oil passage shall be provided in one of the pins through which a lubricant may be applied to the working surfaces (see 4.2.1).

3.3.6 Fastenings. Set screws shall not be used in universal joints (see 4.2.1).

3.4 Performance. The universal joint shall be capable of transmitting the static torque in accordance with table II at the indicated angle without rupture, permanent set, or other evidence of failure (see 4.3.2).

3.4.1 Backlash. The universal joint shall exhibit no more than the maximum backlash in accordance with table IV (see 4.3.1).

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TABLE IV. Maximum backlash.

Hub Outside Diameter (inches)	Backlash (Degree)
⅜ – 1	0.6
1⅛	0.5
1¼	
1½	
1¾	
2	
2¼	0.3
2½	
2¾	
3	
4	

3.4.2 Efficiency. The universal joint shall transmit the torques in accordance with table II with an efficiency of not less than 98 percent with the axes of the hubs at 180 degrees, and 80 percent with the joint at an angle of 150 degrees between the axes of the hubs (see 4.3.2).

3.5 Identification marking. Each joint shall be permanently and legibly marked with the manufacturer's name, trademark, or symbol in accordance with MIL-STD-130 (see 4.2.1).

3.6 Surface roughness. All nonworking-machined surfaces shall not exceed 125 microinch finishes, and all bearing surfaces shall not exceed 32 microinch finishes (see 4.2.1).

3.7 Workmanship. Workmanship shall be in accordance with high grade commercial practice governing this type of material. Each universal joint shall be uniform in quality and temper and shall be free from burrs, silvers, gouges, porosity, cracks, objectionable scale, or any other defects that may adversely affect serviceability (see 4.2.1).

4. VERIFICATION

4.1 Sampling.

4.1.1 Lot. All joints of the same class and size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 Sampling for visual and dimensional examination. A random sample of joints shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.2 Lot acceptance inspection and tests.

4.2.1 Visual and dimensional examination. Each of the sample units selected in accordance with 4.1.2 shall be visually and dimensionally examined to verify compliance with this specification.

4.2.2 Tests. Each of the sample joints selected as specified in 4.1.2 shall be subjected to the tests specified in 4.3 to determine compliance with this specification.

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4.3 Test procedures.

4.3.1 Backlash. The backlash of the joints shall not exceed the amount specified in table IV when the joint is mounted in a test apparatus with the hubs at 180 degrees. The following details shall apply:

a. Apply a torque of 3.5 to 4.5 inch-pounds for joint sizes 3⁄16 through 3⁄4 inch or 7.5 to 8.5 inch-pounds for all other sizes to the universal joint to obtain the initial reading.

b. Reverse the torque and take a second reading.

c. The backlash at the circumference of the universal joint is the difference between the two readings.

4.3.2 Torque. The joint shall be subjected to the torques specified in table II at four or more radial positions of the hub and at the angles specified. The load shall be sustained at each position for 3 minutes.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Universal joints manufactured to this specification are intended for naval service. Universal joints are expected to operate during and following shock loadings and are intended to last for the life cycle of the ship or craft. Class A universal joints are intended for general purpose use in mechanical remote control systems. Class B joints are intended for use where corrosive conditions are abnormal or nonmagnetic materials are required. Class C joints are intended for use where corrosive conditions are abnormal or high torque requirements are desired. Current stock of universal joints may be used until depleted, future requisitions for universal joints should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. PIN and number required (see 1.3).

c. Whether hub should be drilled, and hub dimensions (diameter and length) (see 3.3.2).

d. AQL 1.0 (see 4.1.2).

e. Packaging requirements (see 5.1).

6.3 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website: http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see 3.2).

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6.4 Subject term (key word listing).

Bearing pin

Hub

Torque

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – SH
DLA – CC

Preparing Activity:
Navy – SH
(Project 3010-2019-001)

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